
Zev-7

Nixie Clock

Assembly Manual

April 2026

This manual describes the assembly, testing, and bring-up of the Zev-7 Nixie Clock from bare PCB and components.

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Designed by Mitchell Feig
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1. Introduction

This manual covers the assembly of the Zev-7 Nixie Clock from bare PCB and components. The Zev-7 uses seven Nixie tubes driven by three HV5530 high-voltage shift registers, an ESP32 microcontroller for WiFi and NTP timekeeping, a high-voltage boost converter, and addressable RGB LED backlighting.

The circuit is documented across four schematic sheets reproduced in Appendix B:

- Power Supply (ps.dxf) — 12V input, 5V regulator, 170V HV supply
- Processor (cpu.dxf) — ESP32, level shifter, Schmitt trigger, sensors, buttons
- Display (nixie.dxf) — HV5530 shift chain, Nixie tubes, neon indicator lamps
- RGB LEDs (led.dxf) — APA106 tube backlights, APA102 under-lights

Components should be soldered in the order presented in Section 4, progressing from low-profile parts to tall components, and from power to signal to display.

2. Safety Warnings

WARNING: This circuit generates approximately 170V DC for the Nixie tube anodes. This voltage is lethal. Never touch any part of the high-voltage circuit while the clock is powered on. Always disconnect power and wait at least 30 seconds for capacitors to discharge before working on the board.

WARNING: Nixie tubes are glass and fragile. Handle tubes by the base only. Do not apply excessive force when inserting tubes into sockets.

CAUTION: Observe correct polarity on all electrolytic capacitors (C1, C2, C3), diode D2, and ICs. Reversed polarity can cause immediate component damage.

CAUTION: The LM2576T-5.0 and Tayloredge 1363 module may become hot during operation. Allow adequate ventilation and do not touch metal tabs during operation.

NOTICE: If for any reason the above conditions and the copyright notice at the end of this manual cannot be met, return the kit or assembled clock in original condition within 30 of purchase for a full refund, less shipping.

3. Required Tools and Equipment

Soldering

- Temperature-controlled soldering iron (recommended 350°C / 660°F)
- Fine solder (0.5–0.8 mm rosin-core, 60/40 or lead-free)
- Solder flux (liquid or paste)
- Solder wick or desoldering pump
- Tweezers (fine-tip, for SMD components)
- Magnifying visor or bench magnifier

Test Equipment

- Digital multimeter (must measure DC voltage up to 200V)
- 12V DC power supply (minimum 1A, center-positive)
- Oscilloscope (optional, useful for debugging data signals)

Hand Tools

- Wire strippers and flush cutters
- Anti-static wrist strap

4. Assembly Sequence

Assemble in the order below, progressing from lowest-profile components to tallest, and from power to signal to display. Test each section before proceeding to the next.

4.1 Step 1: Power Supply

Build and test the power supply first to ensure clean voltage rails before installing sensitive ICs.

Install Components

1. F1 (1 A fuse) — Solder the fuse in series with the positive input from J1.
2. C1 (330 μ F 16V) — Input filter capacitor on the 12V rail. Observe polarity: negative stripe to GND.
3. LM2576T-5.0 — 5V switching regulator. Pin 1 (IN) to 12V, Pin 2 (OUT) to L1, Pin 3 (GND) to ground, Pin 5 (ON/OFF) per datasheet (tie low to enable). Solder the tab for heat dissipation.
4. D2 (1N5819) — Schottky catch diode. Cathode to the junction of LM2576 OUT and L1, anode to GND.
5. L1 (100 μ H) — Output inductor. Connects between LM2576 OUT pin and the 5V output rail.
6. C2 (330 μ F 16V) — Output filter capacitor on the 5V rail. Observe polarity.
7. R1 (10k), R22 (39k)
8. J1 (DC barrel jack) — 12V input connector. Center pin positive.

Test the 5V Rail

Connect a 12V supply to J1. Measure the output with a multimeter:

- VCC / VDD rail should read $5.0V \pm 0.1V$
- If voltage is incorrect, check D2 orientation and L1 connections

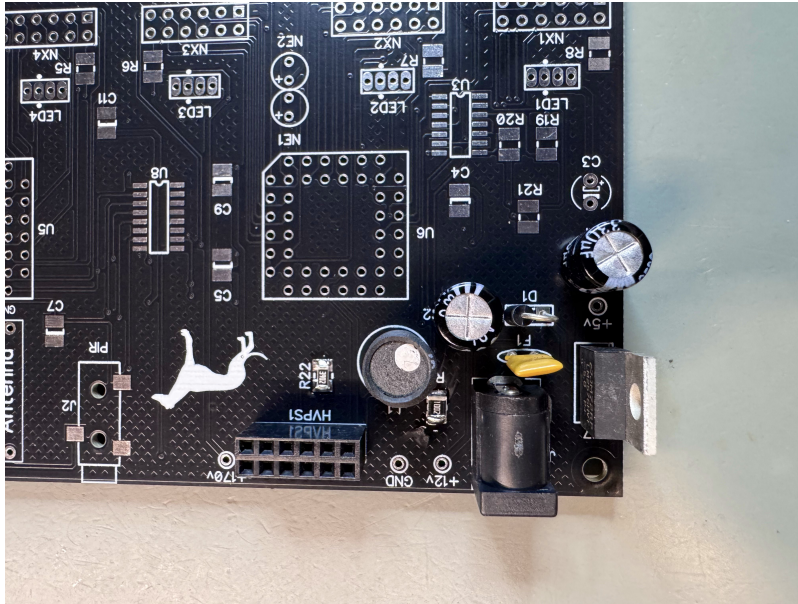
Disconnect power before proceeding.

Install HV Supply

9. HVPS1 (Tayloredge 1363) — HV boost converter module. Temporarily connect a wire between pin 10 and +5v, to enable the power supply. Remove the temporary wire.

Test

Apply 12V power. Check the 170v, 12v, and 5v test points with a meter. The HV supply will read approximately 172v. Remove the temporary wire.



4.2 Step 2: Processor and Support ICs

SMD Components First

Solder all SMD parts before through-hole components.

1. C3 (100 μ F), C4-C11 (100 nF ceramic) — Decoupling and bulk capacitors. Solder these first as they are the lowest-profile components.
2. R14 (39k), R15 (2.7k), R16 (2.7k), R17 (10k), R18 (10k) — R14 is the light sensor load. R15 / R16 are I²C pull-ups to VEE (3.3V). R17 pulls HVENABLE low. R18 provides pull-up for SW2.
3. U3 (74HCT14D, SOIC-14) — Hex Schmitt trigger. Pin 1 dot or notch toward the pin 1 marking on the PCB. Pin 14 to VCC (5V), Pin 7 to GND.
4. U8 (CD4504BM, SOIC-16) — Hex level shifter. Pin 1 to VCC (5V input side), Pin 16 to VDD (5V output side), Pin 8 (VSS) to GND, Pin 13 (SELECT) to VDD.

Through-Hole Components

5. U1 (ESP32 PICO) — Do not solder the module to the board! Cut the single row header strip to fit, and solder in place.
6. Q23 (TEPT4400) — Ambient light phototransistor. Install with the lens facing outward toward ambient light. Install with the shorter lead toward the flat side of the screening.

4.3 Step 3: Display Drivers (HV5530)

1. U4-U6 (HV5530, PLCC-44) — Carefully orient the three sockets. Orient the socket so the flat corner is on the back-left of the board.

WARNING: Do not get this wrong. It is extremely difficult to remove these sockets once soldered, and requires specialized equipment.

4.4 Step 4: Nixie Tubes and Neon Lamps

WARNING: The anode resistors connect to the 170V HV rail. Double-check all connections before applying power. A short on the HV rail can damage components instantly.

Anode Resistors

1. R2–R8 (15k, ×7) — One resistor per Nixie tube. Each connects between the tube's anode (pin 1) and the HV rail (+170V). These limit current to approximately 10–12 mA per tube.

Neon Lamp Resistors

2. R9–R13 (270k, ×5) — One resistor per neon lamp. R9 → NE1 (TLNeon), R10 → NE2 (BLNeon), R11 → NE3 (TMNeon), R12 → NE4 (BMNeon), R13 → NE5 (BRNeon).

Nixie Tube Sockets, Boards, and Tubes

3. Install tube sockets — Solder sockets for NX1–NX7. Verify pin 1 alignment. Do not insert tubes until after the HV voltage has been set and tested (Section 6).

4. Different tube types are available. Each tube type requires a specific tube board. Carefully solder each tube to a tube board, noting the side on which to install the tube. First install the male connector to the tube board, then the tube on the other side.

5. Install neon lamps — Solder NE1–NE5 into position between tube pairs as colon separators.

Tube Position Reference

Tube	Position	Displays	Anode Resistor	Driven By
NX1	Leftmost pair	Hours tens	R8	U6
NX2		Hours ones	R7	U6
NX3	Middle pair	Minutes tens	R6	U6
NX4		Minutes ones	R5	U5
NX5	Rightmost pair	Seconds tens	R4	U5
NX6		Seconds ones	R3	U5
NX7		Tenths of second	R2	U4

Neon Lamp Reference

Lamp	Position	Signal	Resistor	Function
NE1	Between NX1/NX2	TLNeon	R9	Top-left colon
NE2	Between NX1/NX2	BLNeon	R10	Bottom-left colon
NE3	Between NX3/NX4	TMNeon	R11	Top-middle colon
NE4	Between NX3/NX4	BMNeon	R12	Bottom-middle colon

NE5	Near NX7	BRNeon	R13	Right indicator
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4.5 Step 5: RGB LEDs

Chain 1: Tube Backlights (APA106)

1. R19 (150Ω) — Data line series resistor. Connects between the Schmitt trigger output (LEDws, U3 pin 10) and LED1 DIN.
2. LED1–LED7 (APA106) — Seven single-wire addressable RGB LEDs. Daisy-chained: LED1 DOUT → LED2 DIN → ... → LED7 (end of chain). Position one LED behind each Nixie tube. All VCC pins to 5V, all GND pins to ground. Driven from ESP32 IO5.

Chain 2: Under-Lights (APA102)

3. R20 (150Ω), R21 (150Ω) — Series resistors on the APA102 data and clock lines respectively, between the Schmitt trigger outputs (U3) and LED8.
4. LED8–LED11 (APA102) — Four SPI addressable RGB LEDs. Daisy-chained: LED8 DOUT → LED9 DIN → LED10 DIN → LED11 (end of chain). Mounted underneath/behind the clock. Data from ESP32 IO32, clock from IO27. All VCC to 5V, GND to ground.

CAUTION: APA106 and APA102 LEDs have different pinouts. Carefully follow the silkscreen when installing.

4.6 Step 6: Sensors

BME280 Temperature/Pressure/Humidity Sensor

1. J3 (4-pin header) — I²C sensor connector. Pin 1 = VEE (3.3V), Pin 2 = GND, Pin 3 = SCL, Pin 4 = SDA. The BME280 module plugs into this header. I²C pull-ups R15 / R16 (2.7k) are already installed.

The BME280 is auto-detected at I²C addresses 0x76 and 0x77. The sensor is optional — the clock operates normally without it.

PIR Motion Sensor

2. J2 (PIR connector) — Connector for the PIR motion sensor module. Signal line connects to PIR net (ESP32 IO37). Decoupled by C7 (100 nF).

Light Sensor

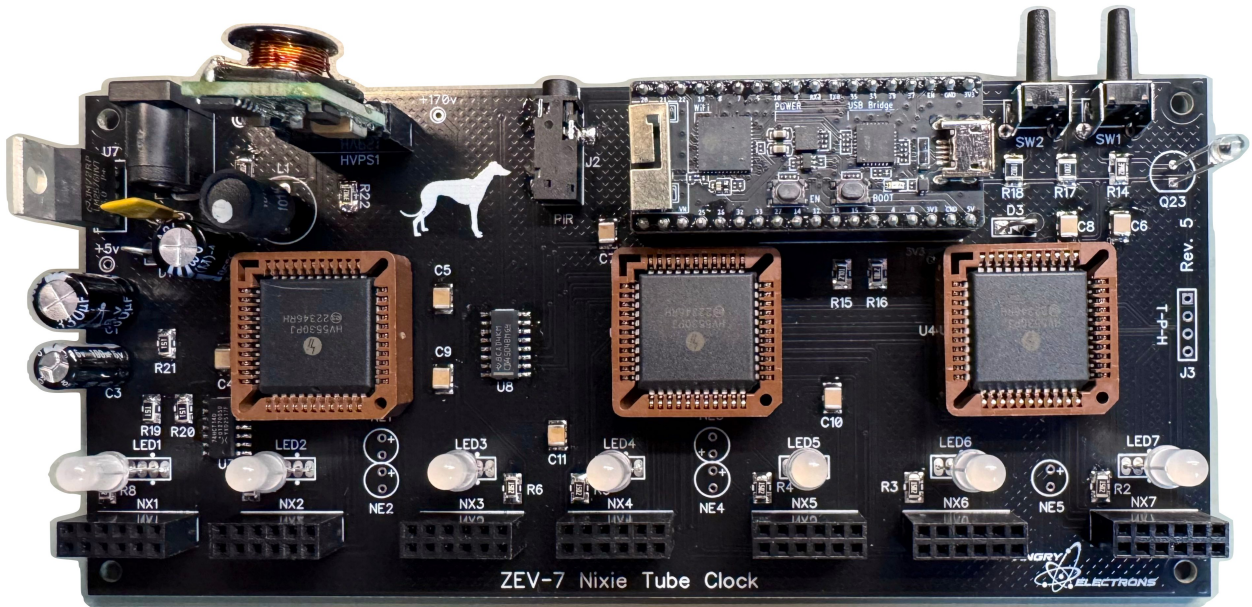
3. Q23 (TEPT4400) — Ambient light phototransistor. Install with the lens facing outward toward ambient light. Install with the shorter lead toward the flat side of the screening.

4.7 Step 7: Buttons and Connectors

1. SW1 (Pushbutton) — Connects to ESP32 IO34 (input-only pin). Active-low with external pull-up.

2. SW2 (Pushbutton) — Connects to ESP32 IO35 (input-only pin). Active-low with external pull-up (R18).

The Completed Board - Without Colons Installed



The Completed Board - Without Colons Installed

5. Initial Power-On and Testing

WARNING: Do not insert Nixie tubes until the HV rail has been verified.

Pre-Power Checks

Before applying power for the first time:

- Visually inspect all solder joints under magnification
- Check for solder bridges, especially on SOIC and PLCC packages
- Verify all electrolytic capacitor polarities (C1, C2, C3)
- Verify diode orientations (D2, D3)
- Confirm no shorts between power rails: measure resistance between VCC and GND, HV and GND

Power-On Sequence

1. Connect 12V supply to J1. Current draw should be under 100 mA with no tubes installed.
2. Verify voltages:

- VCC / VDD: $5.0V \pm 0.1V$
 - VEE (ESP32 3.3V rail): $3.3V \pm 0.1V$
 - HV rail: 170V (no adjustment needed)
3. The ESP32 should create a WiFi access point named "Zev-7" on first boot.
 4. Connect from a cell phone or PC to the access point a browser to 192.168.4.1. At a minimum select the network and enter the network password. Scroll to the bottom of the form and press Save. Press Back, and then Start, on the home page.
 5. After WiFi connects, the display shows the startup sequence, and starts running.

Testing Subsystems

LEDs: After WiFi connection and time sync, the RGB backlights should activate with the default "Random Multi" effect. If LEDs do not light, check R19, R20, R21, U3, and data line continuity.

Buttons: Press SW1 — should trigger a button override. Hold SW2 for 5 seconds — should restart in AP mode.

Sensors: If BME280 is connected, enable it in Setup. Temperature, humidity, and pressure appear in the info display cycle. Triple-click SW2 to verify the light sensor (raw reading on NX1–NX2).

6. Troubleshooting

Symptom	Check	Solution
No power (nothing lights)	F1 fuse, J1 connection	Replace fuse. Check barrel jack center-positive.
5V rail incorrect	LM2576 connections, D2 orientation, L1	Verify LM2576 pinout. Check D2 Schottky cathode direction. Confirm L1 value.
ESP32 not booting	VEE (3.3V), solder joints on module	Reflow ESP32 module pins. Check for shorts on IO0 (must be high for normal boot).
LEDs not lighting	R19, R20, R21, U3 (74HCT14D), VCC to LEDs	Verify 5V at LED VCC pins. Check R19 / R20 / R21 continuity. Verify U3 pin 14 = 5V.
LED chain partially works	Failed LED in chain	Data cascades: a dead LED breaks the chain. Replace the first non-working LED.
No display (tubes dark)	HVENABLE, HV rail, R2–R8	Verify HVENABLE is high when display should be on. Check HV voltage. Verify anode resistors.
Wrong digits displayed	DATA / CLK / LE wiring, U8 level shifter	Verify U8 connections. Check shift chain order (U6→U5→U4). Confirm CLK / LE / DATA signals with scope.
Digits dim or flickering	HV voltage, BL (blanking) signal	Verify HV rail at 170V. Verify BL PWM signal from IO13 through U8.
Neons not working	R9–R13, HV5530 outputs for neons	Verify 270k resistors. Check that neon outputs toggle (use scope on HV5530 output pins).
No WiFi AP on first boot	ESP32 flash, antenna	Reflash firmware via USB. Ensure no metal near ESP32 antenna area.
BME280 not detected	J3 wiring, I ² C pull-ups R15 / R16	Verify SDA / SCL at J3. Check R15 / R16 to 3.3V. Try alternate I ² C address module.
PIR not responding	J2 wiring, C7, IO37	Verify signal at IO37. Check PIR module power (3.3V). Confirm

		C7 decoupling.
Light sensor not reading	Q23 orientation, R14	TEPT4400 must face ambient light. Verify R14 (39k) and signal at IO38.

Appendix A — Bill of Materials

Component values reflect the latest schematic revisions in Appendix B. Reference designators match the silkscreen on the PCB.

Integrated Circuits

Ref	Part	Package	Qty	Description
U1	ESP32 PICO KIT V3	Module	1	ESP32-PICO-D4 WiFi microcontroller
U3	74HCT14D	SOIC-14	1	Hex inverting Schmitt trigger
U4	HV5530	PLCC-44	1	HV shift register (NX7 + BR neon)
U5	HV5530	PLCC-44	1	HV shift register (NX4, NX5, NX6 + TM/BM neons)
U6	HV5530	PLCC-44	1	HV shift register (NX1, NX2, NX3 + TL/BL neons)
U8	CD4504BM	SOIC-16	1	CMOS hex voltage level shifter
—	LM2576T-5.0	TO-263-5	1	5V 3A switching regulator
HVPS1	Tayloredge 1363	Module	1	HV boost converter (~170V)

Resistors

Ref	Value	Qty	Description
R1	10k	1	HV voltage divider
R2–R8	15k	7	Nixie tube anode current limiters
R9–R13	270k	5	Neon lamp current

			limiters
R14	39k	1	Light sensor (TEPT4400) load resistor
R15, R16	2.7k	2	I ² C pull-ups (SDA, SCL)
R17	10k	1	HVENABLE pull-down
R18	10k	1	Button pull-up (SW2)
R19	150Ω	1	APA106 data line series resistor
R20	150Ω	1	APA102 data line series resistor
R21	150Ω	1	APA102 clock line series resistor
R22	39k	1	39k resistor

Capacitors

Ref	Value	Qty	Description
C1	330 μF 16V	1	Input filter (12V rail), electrolytic
C2	330 μF 16V	1	Output filter (5V rail), electrolytic
C3	100 μF	1	ESP32 supply bulk capacitor
C4–C11	100 nF	8	Decoupling capacitors (ceramic)

Diodes

Ref	Part	Qty	Description
D2	1N5819	1	Schottky catch diode (LM2576)
D3	Protection diode	1	VCC protection (near ESP32)

Inductors and Fuses

Ref	Value	Qty	Description
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L1	100 μ H	1	LM2576 buck converter output inductor
F1	1 A fuse	1	Input power fuse

Nixie Tubes and Neon Lamps

Ref	Part	Qty	Description
NX1–NX7	Nixie tube	7	Numeric display tubes (digits 0–9)
NE1–NE5	Neon indicator lamp	5	Colon and status indicators
	12 Pin Male Socket	7	Mounts to PC Board
	12 Pin Female Socket	7	Mounts to the bottom of the tube boards

LEDs

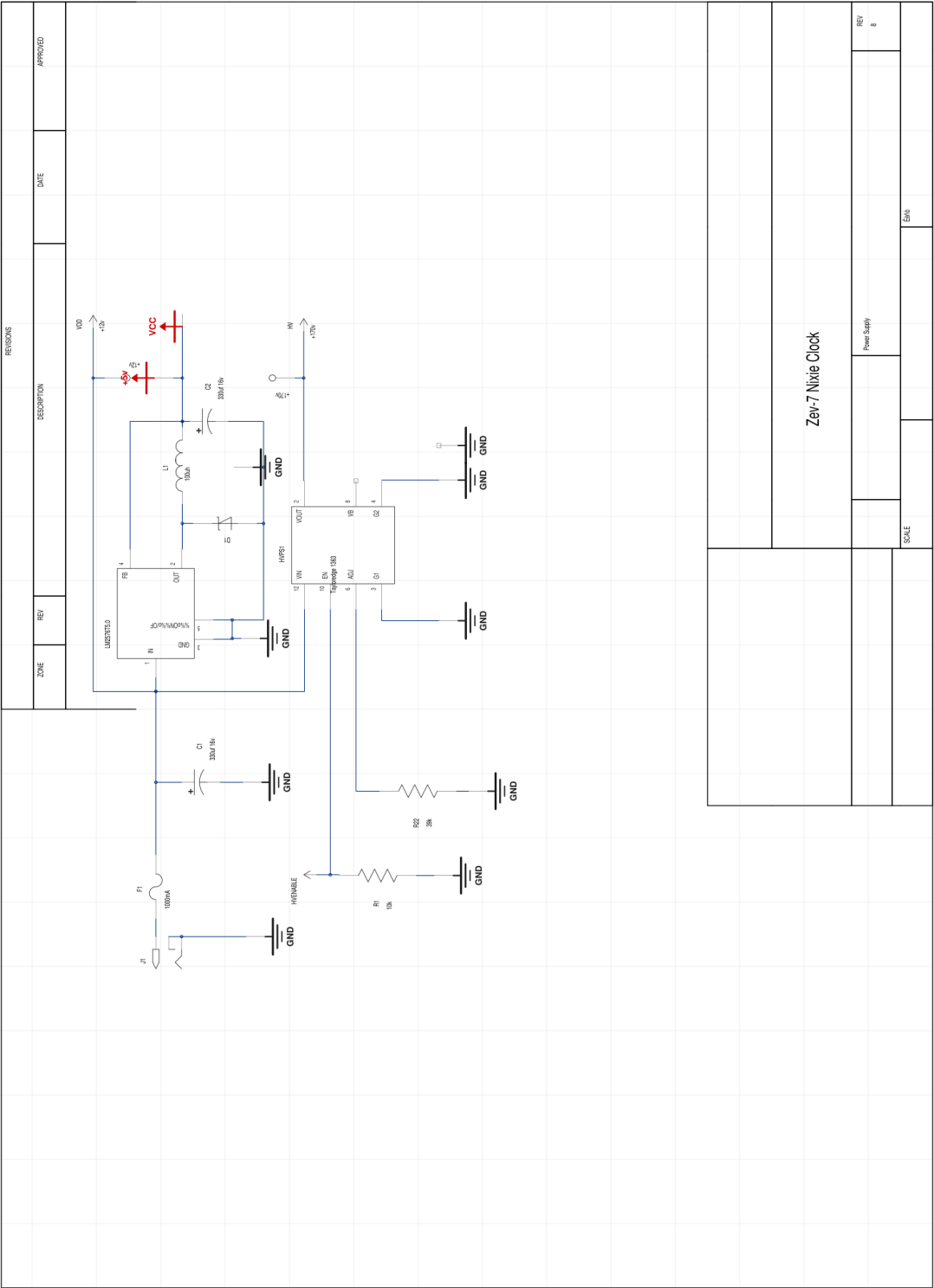
Ref	Part	Qty	Description
LED1–LED7	APA106 RGB LED	7	Tube backlight (single-wire, IO32)
LED8–LED11	APA102 RGB LED	4	Under-light (SPI, IO32 / IO27)

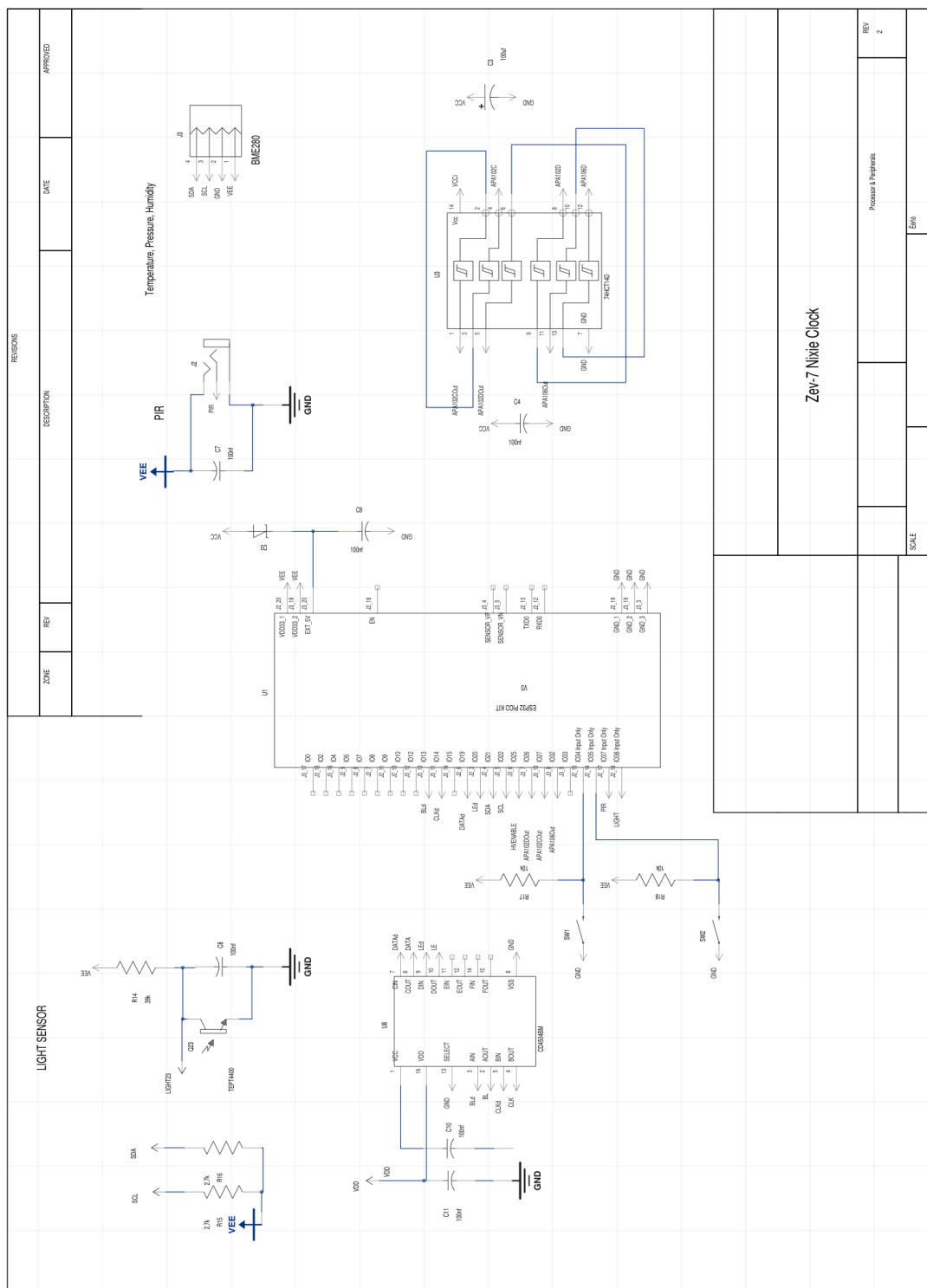
Connectors, Switches, and Sensors

Ref	Part	Qty	Description
J1	DC barrel jack	1	12V power input
J2	PIR connector	1	PIR motion sensor header
J3	4-pin header	1	BME280 I ² C sensor connector
SW1	Pushbutton	1	Button 1 (override, IP, OTA)
SW2	Pushbutton	1	Button 2 (calibration, AP mode)
Q23	TEPT4400	1	Ambient light phototransistor

Appendix B — Schematics

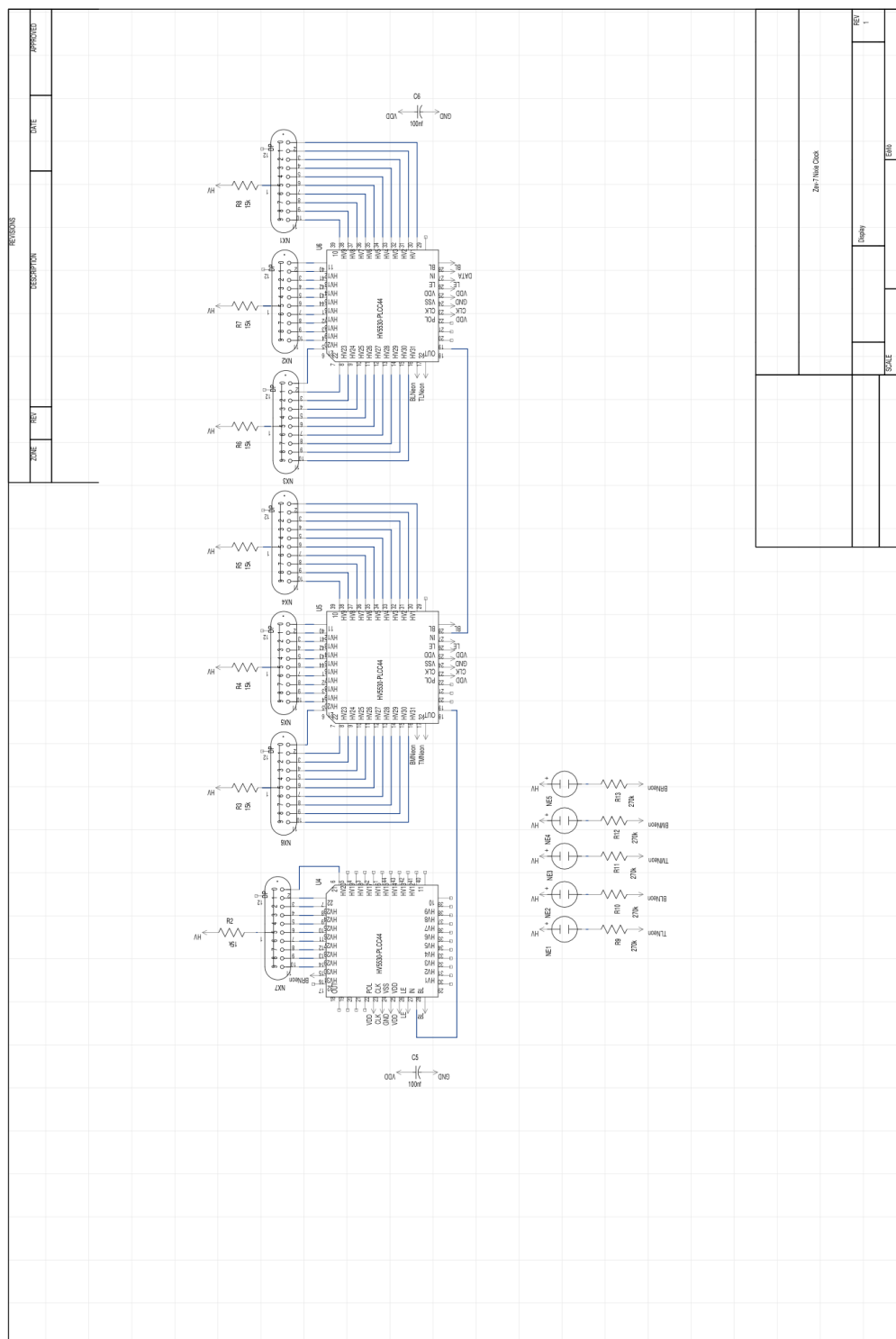
Each of the four schematic sheets is reproduced in landscape orientation on its own page on the pages that follow. Sheets are presented in the assembly order: Power Supply, Processor, Display, RGB LEDs.

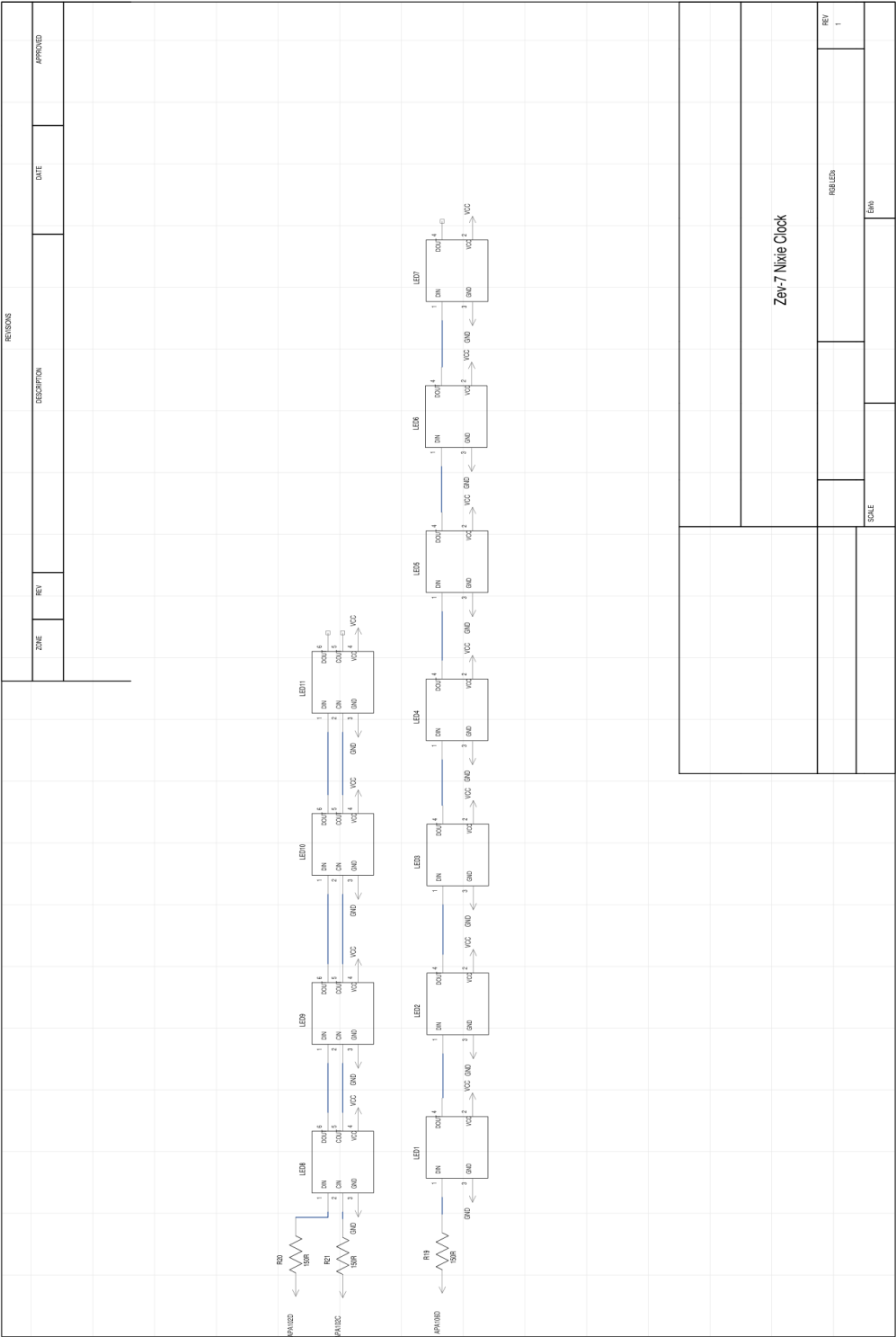


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Processor

Sheet 1 of 1





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The Nixie tubes used in this clock contain a small amount of mercury and neon gas under low pressure. The high-voltage circuitry inside generates approximately 170 volts and should not be touched while the clock is powered on.

Zev-7 Nixie Clock — Designed by Mitchell Feig